
Release: September 25, 2026.

Official deadline: Friday, October 9, 2026, at 11:59pm ET.

0% penalty extension: Submissions are accepted through Monday, October 12, 2026, at 11:59pm ET with no late penalty. No extension request is needed.

Final-report planning date: December 4 (provisional).

Overview

The research project gives you the opportunity to investigate a research question of your choosing in depth. You will identify a question, situate it within the existing literature, propose and carry out an approach, and report your findings.

Your project must relate to **topics covered in ECE750T**, including topics scheduled for later in the term. These include algorithmic fairness, evaluation and auditing, measurement and data curation, privacy, predictive multiplicity, data attribution, machine unlearning, foundation-model safety, data poisoning and collective action, and AI governance. State the connection explicitly in your proposal.

This is an **individual project**. Work independently; you are welcome to discuss ideas at a high level with classmates.

This short project is meant to give you experience with academic research. You are not expected to set a new state of the art. The main point is to demonstrate a sound research process, regardless of the outcome. A well-designed small study with a clear question is sufficient. The final report will be **4 pages, excluding references and appendices**; this handout concerns the proposal.

Proposal submission

- **Format:** A PDF of at most **2 pages, excluding references**, using the supplied LaTeX template (`project-proposal.tex` in `proposal-template.zip`). Retain the eight question headings and replace the placeholder text.
- **Content:** Address all eight questions below. Some answers may be preliminary. The proposal should demonstrate a clear question, an initial literature review, and a feasible plan.
- **Submission:** Submit the same single PDF to **both Crowdmark and LEARN Dropbox**.

Research-planning questions

The proposal follows the adapted Heilmeier Catechism used in ECE657D: eight questions for evaluating a research plan.¹ The provided template has a section for each question.

[3 marks] Q1: What is your research question?

State the question in plain language. What are you trying to accomplish? A reader without a background in your specific topic should understand what you are investigating. Explain how the question relates to one or more topics covered in ECE750T.

[3 marks] Q2: What prior work will you compare against, and what are its limitations?

Summarize the most relevant existing approaches. What has been tried before? What limitations or gaps motivate your work? Cite specific papers.

¹The original [Heilmeier Catechism](#), developed by George Heilmeier, is available from DARPA.

[3 marks] Q3: What is novel about your proposed approach?

Explain what is new or different. This need not be a major research contribution: it could be a new combination of existing techniques, an application to a new domain, or a systematic comparison that has not been done before.

[2 marks] Q4: Why does this matter? What is the potential impact?

Who would benefit from your results? Why should we care about the question? Connect your work to broader goals in responsible AI or its applications.

[2 marks] Q5: What could go wrong? What are the key assumptions?

Identify the main risks and assumptions. What are the possible failure modes? Are there assumptions about the data, compute, method, or setting that might not hold?

[2 marks] Q6: What compute and data resources do you need, and how will you obtain them?

Describe your resource plan. What datasets or other materials will you use? What compute do you need, and where will you get it? Be specific about the access you have and what you still need to arrange. For a theoretical or analytical project, explain the resources and evidence your method requires.

[2 marks] Q7: What is your week-by-week plan?

Break the period between the proposal and final submission into concrete weekly milestones. Using the draft dates above, plan for approximately eight weeks, including reading week. What will you accomplish each week? Allow time for checking results and writing the report.

[2 marks] Q8: What are your milestones, and how will you measure success?

Define the quantitative and/or qualitative criteria you will use to evaluate your approach. What constitutes a successful project? What baselines or other comparisons will you use? Your criteria should make it possible to learn from negative results as well as positive ones.

Scope and resources

Match the scope to the time, data, and compute you can access. A well-designed small-scale experiment is preferable to an ambitious but infeasible large-scale one. Smaller models and datasets can make CPU-based experiments practical. If you plan to use shared or external compute, explain how you will obtain access and what you can complete if it is unavailable.

You will not be penalized for lacking external GPU resources. You are expected to demonstrate a sound research process with the resources available to you.

Grading

The proposal is worth **20 marks**: 19 marks across Q1–Q8 and **1 mark for writing quality, citations, and compliance with the format**. Preliminary answers are acceptable when the remaining uncertainty and a plan to resolve it are clear.

Use of AI tools

The ECE750T course policy permits generative AI as a research and coding aid for the project, with use documented in the final report. Include a brief disclosure in your proposal of any tools used and for what purpose. You are responsible for checking the material you submit and being able to explain your proposed work.